

CO-CURRICULAR ACTIVITY

FPGA Development Board

Lattice ECP5 Hardware Platform

Bi-Weekly Progress Report – 01

Project Team

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Branch: Electronics and Communication Engineering

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1 Project Objective

The objective of the project is to support the operation of the ECP5 FPGA device by providing a complete development platform for it. Our current objective is to identify the requirements of the ECP5 module and to design a system-level architecture of what needs to be achieved and how. The project will be developed in a modular manner, with each subsystem being designed independently before being integrated into a complete system.

2 System Architecture Overview

To support the ECP5 FPGA effectively, the system has been partitioned into six separate subsystems:

1. Subsystem 1: Power Subsystem
2. Subsystem 2: Configuration and Programming Interface
3. Subsystem 3: PCB Layout and Mechanical Constraints
4. Subsystem 4: Clocking and Timing Network
5. Subsystem 5: User I/O and Peripheral Interfaces
6. Subsystem 6: Debugging and Telemetry Infrastructure

The exact implementation of these subsystems will depend on the selected ECP5 device/package, the required peripherals, the available board area, and other project requirements.

The reference architecture studied for initial baseline requirements is the Lattice ECP5-5G Evaluation Board, adapted to meet the peripheral constraints of this project. The purpose of studying the evaluation board is not to reproduce the design directly, but to use it as a practical reference for identifying the types of supporting circuits that may be required. The final board architecture will be adapted according to the requirements and constraints of the present project.

3 Subsystem Progress Reports

3.1 Subsystem 1: Power Subsystem

Lead: *Pranjal Pandey (SC25B128)*

3.1.1 Overview and Design Strategy

The work of the power subsystem is to provide stable and low noise rails to all the different subsystems. To achieve this the current approach is to identify all the required power end points, determine their electrical requirements, and subsequently work backwards towards the selection of regulators and thus a complete power architecture. From the preliminary investigation of the ECP5 requirements and the reference evaluation board, the required electrical domains are categorized in Table 1.

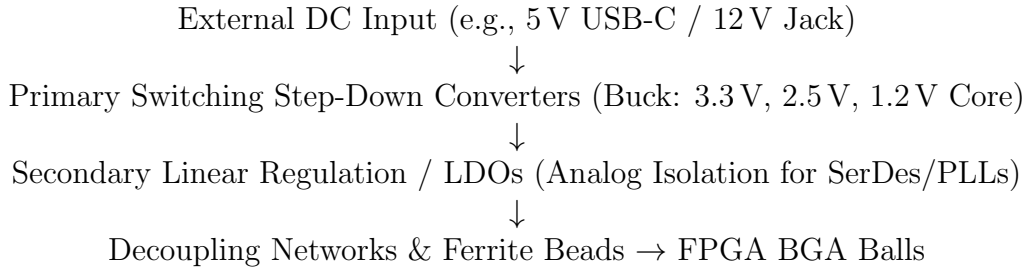
Rail Name	Target Domain	Nominal Voltage & Specifications
V_{CC}	FPGA Digital Core Logic	1.1 V (ECP5) / 1.2 V (ECP5-5G)
V_{CCAUX}	Auxiliary / Logic Buffers	2.5 V ($\pm 5\%$, ramp ≤ 30 mV/ μ s)
V_{CCIO}	I/O Bank Drivers	Configurable: 1.2 V, 1.5 V, 1.8 V, 2.5 V, 3.3 V
V_{CCA}	SerDes Analog Core (if used)	1.1 V (ECP5) / 1.2 V (ECP5-5G)
V_{CCHTX}	SerDes TX Buffers (if used)	1.1 V (ECP5) / 1.2 V (ECP5-5G)
V_{PERIPH}	External Oscillators / Flash	3.3 V / 2.5 V depending on component specs

Table 1: Preliminary power-domain specification and allocation

The exact number of V_{CCIO} rails required along with the voltage levels will depend on the selected FPGA package and the allocation of I/O banks to different interfaces, making FPGA package selection and I/O planning an important dependency for the final power architecture. Similarly, the external peripheral rail (V_{PERIPH}) will depend on the specific peripherals chosen and will have to wait until peripheral selection is complete.

3.1.2 Power Distribution Architecture

The planned power-delivery pipeline follows a hierarchical tree:



For each required rail, the following parameters will be determined:

1. Required voltage
2. Estimated current requirement
3. Expected peak/transient requirement
4. Source/input voltage
5. Regulator topology
6. Decoupling requirements
7. Monitoring and test points

Once these requirements are established, suitable regulator and supporting components can be selected and incorporated into the BOM.

3.1.3 Next Steps for the Power Subsystem

- Establish power sequencing requirements (V_{CCIO} ramp relative to V_{CC} and V_{CCAUX} to ensure clean Power-On Reset). While it has already been mentioned that the ECP5 requires specific ramp rates however the architecture to provide this is yet to be determined. This will be done by studying the evaluation board and the ECP5 datasheet and taking inspiration from the power sequencing of other similar FPGA devices.
- Select regulator ICs based on thermal dissipation, PCB area, and efficiency targets. Currently the plan is to evaluate the ICs already being used in the reference evaluation board and identify whether they are the optimal solution for our design. If not we will be looking for alternatives.

3.2 Subsystem 2: JTAG and FLASH Configuration

Lead: Arnav Kavali (*SC25B120*)

3.2.1 Overview and Design Strategy

We need a reliable way to wake up our main chip, the Lattice ECP5, and load its programming (the bitstream) into its memory when it turns on. To do this, our board needs a USB-to-JTAG bridge and an SPI Flash memory chip.

The ECP5 supports both Master SPI (MSPI) and JTAG boot modes. Our main goal here is getting the JTAG interface working so we can test and debug live from a PC, and setting up the SPI Flash so the board can boot up on its own without being plugged into a computer.

3.2.2 Configuration Architecture

The proposed configuration pipeline includes:

- **JTAG Download Interface:** A built-in download controller will be utilized to convert USB to JTAG, allowing the host PC to directly interface with the ECP5 device. This setup allows device programming using *to be decided*.
- **SPI Flash Memory:** An onboard SPI Flash memory device will store the default bitstream. This ensures the ECP5 can boot automatically upon power-up by reading the bitstream over the SPI bus.
- **JTAG to MSPI Pass-through:** The host will be able to flash the onboard SPI memory indirectly by piping data through the FTDI bridge and the FPGA's JTAG system chain.

3.2.3 Next Steps for JTAG and FLASH

- To confirm with requirements and finalize FT2232H FTDI chip and map out the USB 2.0 routing.
- Select a compatible SPI Flash memory component with adequate density to hold our custom uncompressed bitstream format.

- Map the specific custom configuration of the 144 (assumed) pins (e.g., PROGRAM, INIT, DONE, and boot-mode select switches) to onboard test points and status LEDs for hardware bring-up.

4 Conclusion

The first stage of the project has been focused on converting the general requirement of developing an ECP5-based FPGA board into a set of identifiable engineering problems. The power subsystem has been selected as the initial area of detailed work: the current approach is to identify all required power endpoints, determine their electrical requirements, and subsequently work backwards toward regulator selection and a complete power architecture.

Further work will integrate this subsystem with FPGA configuration, clocking, I/O, debugging, and PCB layout to produce the complete development-board design.